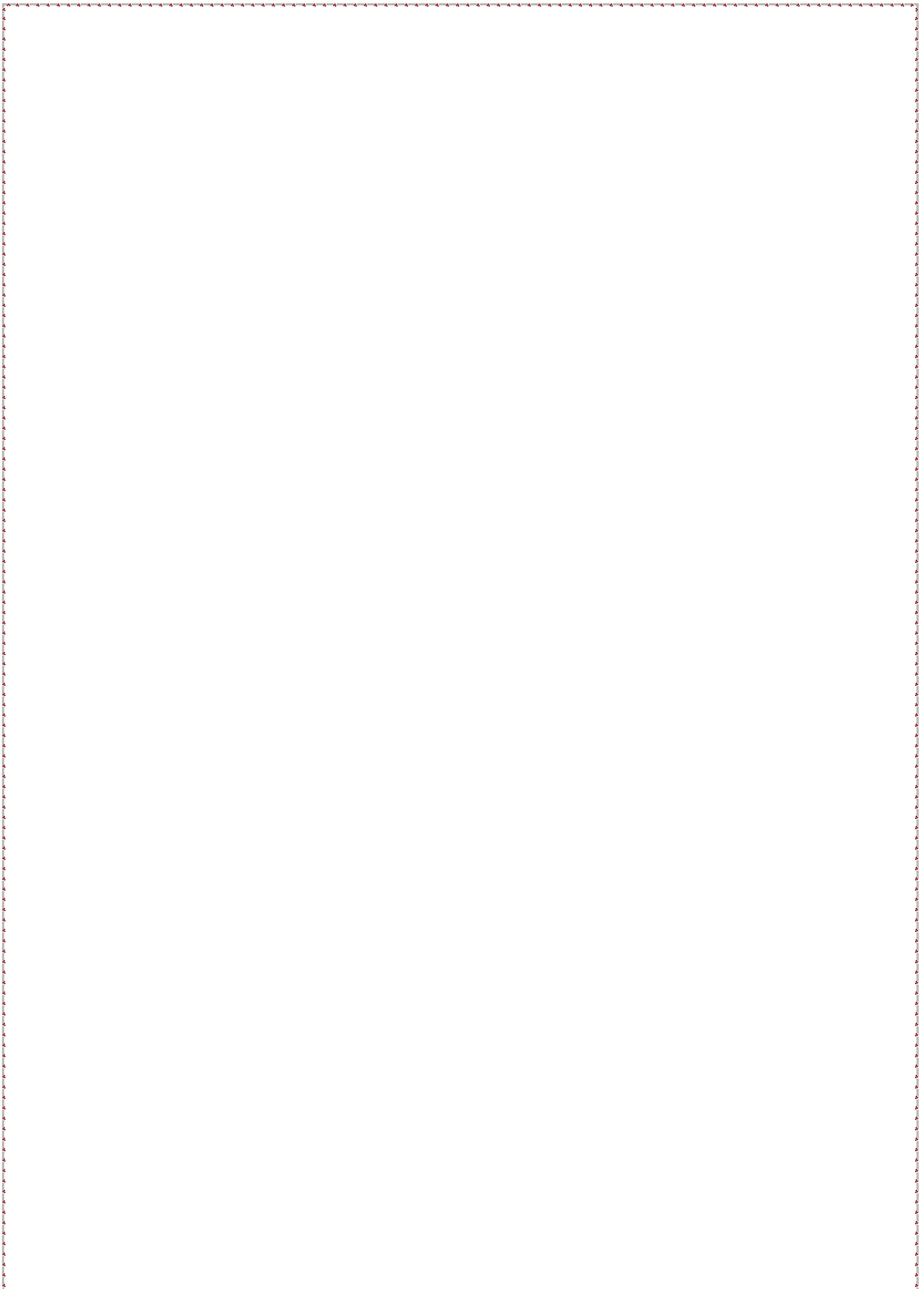


Chapter One



Energy Reserves and Potential





CHAPTER 1

Energy Reserves and Potential

Introduction

Energy reserves are part of the energy resources that, based on technical, economic and other relevant (e.g. environmental) considerations, can be recovered and for which extraction is justified. The exact definition of reserves depends on the kind resources in focus.

Global Classification of Energy Reserves

The classification of energy reserves is guided by the **United Nations Framework Classification for Fossil Energy and Mineral Reserves and Resources (UNFC 2009)**. This framework evaluates resources based on:

- **Economic and Social Viability**
- **Field Project Status and Feasibility**
- **Geological Knowledge**

The **System of Environmental Economic Accounting (SEEA)-Energy** groups the detailed categories of UNFC into three aggregated classes characterizing the commercial recoverability of the resources as follows:

- **Class A:** Commercially recoverable resources.
- **Class B:** Potentially commercially recoverable resources.
- **Class C:** Non-commercial and other known deposits.

Energy Reserves in India

India has a diverse range of energy resources, with significant reserves of both fossil fuels and renewable energy potential. The country's energy mix is undergoing a transformation to reduce dependence on fossil fuels and promote the use of renewable energy sources, driven by environmental, economic, and macroeconomic consideration.

This chapter highlights the energy reserves and potential in India, focusing on both fossil fuels and renewable energy resources. It explores coal, lignite, crude oil, natural gas, and renewable energy potential, providing an overview of the country's energy landscape.

Chapter 1: Energy Reserves and Potential

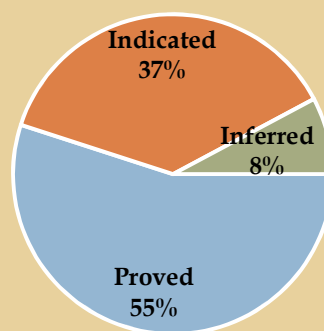
Highlights of reserves in India

1.1 Coal Reserves

India is home to one of the **largest proven coal reserves** in the world. As of **01-04-2025**, the total estimated coal reserves were **400.7 billion tonnes**, showing an addition of **11.29 billion tonnes** compared to the previous year. This marks a **growth of 2.90%** in total coal reserves during the year **2024-25(P)** over the previous year (2023-24) (Table 1.1). The distribution of coal reserves in India is concentrated in a few states like **Odisha, Jharkhand** and **Chhattisgarh**. These three states account for approximately **70% of the total coal reserves** in the country.

Out of the total reserves in the country, proven reserves i.e. economically viable and geologically confirmed account for approximately **55%** of the total estimated coal reserves as depicted in the **figure 1.1**.

Fig 1.1 Estimated Reserves of Coal in India as on 01.04.2025

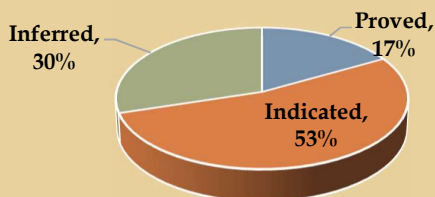


Total Estimated reserves = 400.715 Billion Tonnes

1.2 Lignite Reserves

The total estimated reserves of **lignite** as on **01-04-2025** stood at **47.371 billion tonnes**, showing an increase of **0.07 billion tonnes** over the previous year. This marks an **increase of 0.15%** in lignite reserves during the year **2024-25(P)** compared to **2023-24** (Table 1.2). The highest reserves of lignite are located in the state of **Tamil Nadu (79%)**.

Fig 1.2 Estimated Reserves of Lignite in India as on 01.04.2025



Total Estimated reserves = 47.371 Billion Tonnes

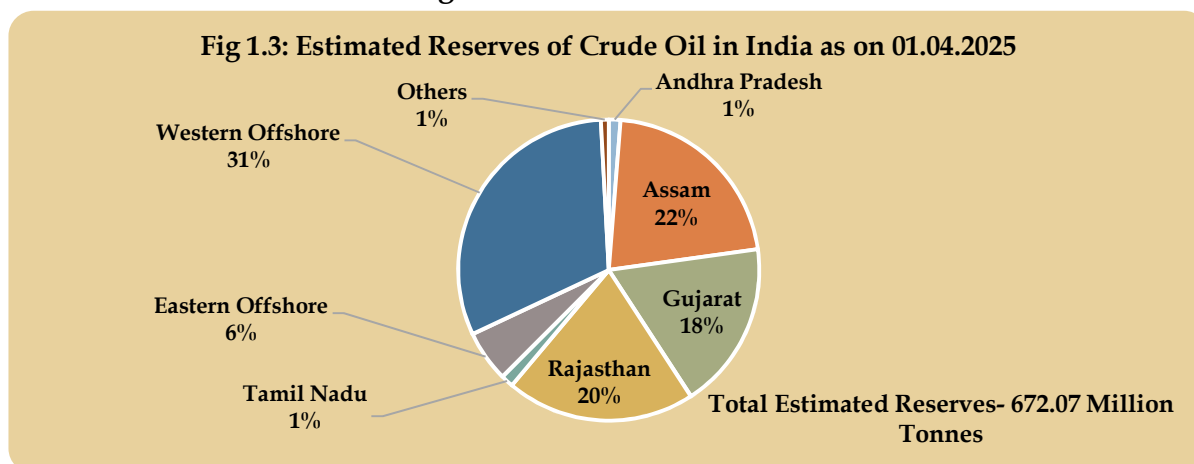
Out of the total lignite reserves in the country, proven reserves of lignite account for approximately **17%** of the total lignite reserves in India as shown in the **figure 1.2**.

** Brief about the Proved, Inferred and Indicated has been given under Annexure-I.*

Chapter 1: Energy Reserves and Potential

1.3 Crude Oil Reserves

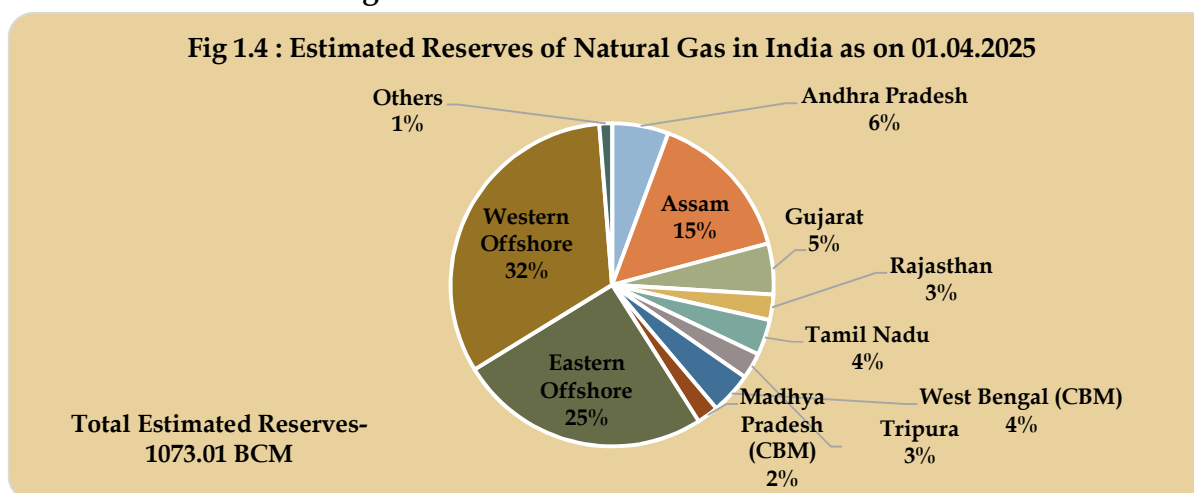
As of **01-04-2025**, the estimated reserves of **crude oil** in India stood at **672.07 million tonnes**, compared to **671.40 million tonnes** in the previous year, reflecting a **0.10% increase** over the last year (Table 1.3). Geographically, the maximum crude oil reserves in India are concentrated in the **Western Offshore** region, which accounts for **31%** of the total crude oil reserves. This is followed by the **Assam** region, which holds around **22%** of the country's crude oil reserves as shown in **figure 1.3**.



Others include Arunachal Pradesh (0.43%), Nagaland (0.35%), Tripura (0.01%), West Bengal (0.01%)

1.4 Natural Gas Reserves

The estimated reserves of **natural gas** as of **01-04-2025** were **1,073.01 Billion Cubic Meters (BCM)** compared to **1,094.19 Billion Cubic Meters (BCM)** in the previous year, reflecting a **decrease of 1.94%** over the last year (Table 1.4). The largest reserves of natural gas in India are located in the **Western Offshore** region, which holds approximately **32%** of the total natural gas reserves. This is followed by the **Eastern Offshore**, which accounts for **25%** of the reserves as shown in **figure 1.4**.



#Others include Arunachal Pradesh (0.67%), Jharkhand (CBM) (0.44%), Nagaland (0.009%)

Chapter 1: Energy Reserves and Potential

1.5 Renewable Energy Potential in India

India is increasingly focusing on renewable energy sources as part of its sustainable development and mitigating impact on climate. As of **March 31, 2025**, the total estimated potential for renewable power generation in India stood at **47,04,043 MW**. This potential comes from a variety of renewable sources, including **wind, solar, biomass, small hydro, and cogeneration from bagasse** (Table 1.4).

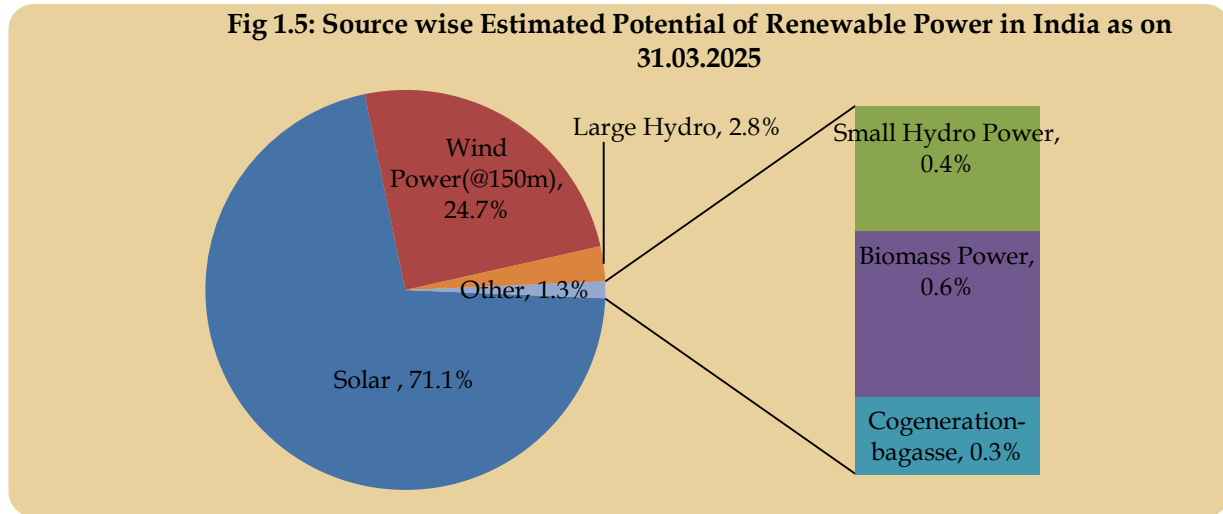


Figure 1.5 shows the potential is spread across several key renewable energy sources, as detailed below:

- **Solar Power:** India has immense potential for solar power generation, estimated at **33,43,378 MW**, which accounts for **71%** of the total renewable energy potential. Solar power stands as the most significant source of renewable energy in India.
- **Wind Power:** With an estimated potential of **1,163,856 MW** (close to 25% of total renewable energy potential), it has the potential to contribute significantly in shaping the renewable energy-generation scenario of India.
- **Large Hydropower (Large Hydro):** The potential for large hydropower generation is estimated at **133,410 MW**, contributing **2.84%** to the country's renewable energy potential.
- **Small Hydro Power (SHP):** India also has a notable potential of **21,134 MW** (0.45%) from **small hydro power** projects.
- **Biomass Power:** Biomass, which includes agricultural waste, forest residues and other organic matter, has a potential of **28,447 MW**, accounting for **0.60%** of the total renewable power potential.
- **Cogeneration from Bagasse:** India has a specific potential of **13,818 MW** (0.29%) from **bagasse-based cogeneration** in sugar mills. This is a highly efficient form of energy generation, especially in regions with a robust sugar industry.

Chapter 1: Energy Reserves and Potential

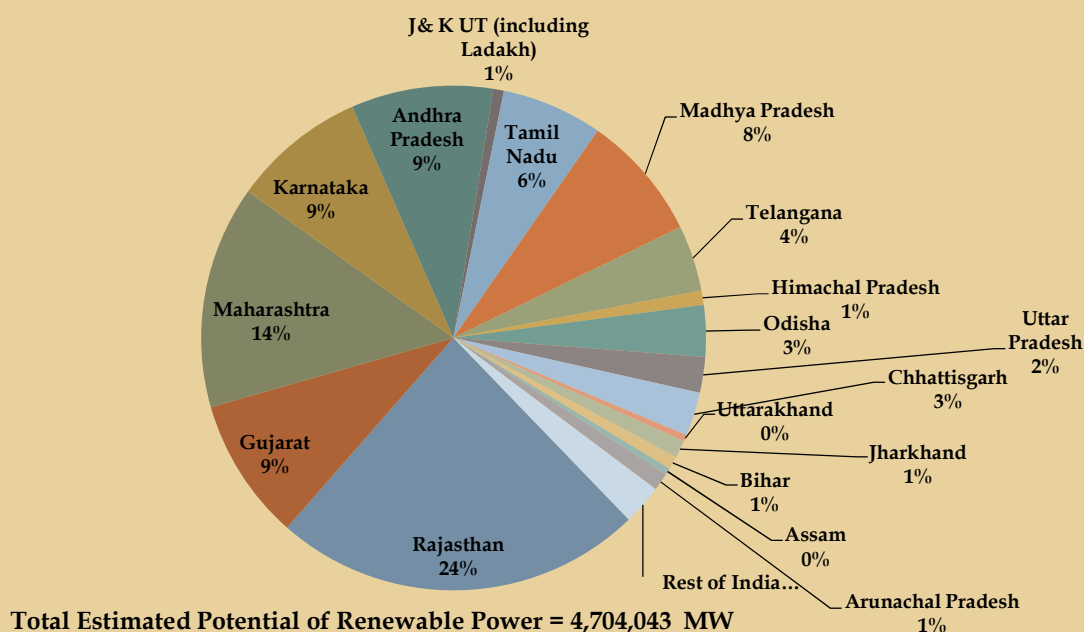
Geographical Distribution of Renewable Energy Potential

The geographical distribution of renewable energy potential in India (**Figure 1.6**) shows significant regional variations. The states with the highest renewable energy potential, as of **March 31, 2025**, are:

- **Rajasthan:** 24% of total potential (11,14,794 MW)
- **Maharashtra:** 14% of total potential (6,71,024 MW)
- **Gujarat:** 9% of total potential (4,27,954 MW)
- **Andhra Pradesh:** 9% of total potential (4,27,933 MW)
- **Karnataka:** 8.6 % of total potential (4,04,227 MW)

Together, these five states account for around 65% of India's total renewable energy potential, highlighting their critical role in India's renewable energy future.

Fig 1.6: Statewise Estimated Potential of Renewable Power (in percentage) in India as on 31.03.2025



Chapter 1: Energy Reserves and Potential

Table 1.1: Statewise Estimated Reserves of Coal (As on 1st April 2025)

(in Million Tonnes)

States/ UTs	Proved		Indicated		Inferred		Total		Distribution (%)	
	2024	2025	2024	2025	2024	2025	2024	2025	2024	2025
Andhra Pradesh	1,025	1,025	2,369	2,369	778	778	4,172	4,172	1.07	1.04
Arunachal Pradesh	31	62	40	23	19	6	90	91	0.02	0.02
Assam	465	465	57	57	3	3	525	525	0.13	0.13
Bihar	2,346	2,346	3,015	6,970	37	37	5,398	9,353	1.39	2.33
Chhattisgarh	40,078	45,463	41,093	38,340	1,495	1,460	82,666	85,263	21.23	21.28
Jharkhand	59,877	60,290	27,135	27,122	4,799	5,842	91,812	93,254	23.58	23.27
Madhya Pradesh	15,425	16,414	12,379	11,796	5,011	5,354	32,815	33,564	8.43	8.38
Maharashtra	8,163	8,339	3,372	3,265	1,817	1,984	13,352	13,587	3.43	3.39
Meghalaya	96	97	17	17	471	470	583	583	0.15	0.15
Nagaland	9	9	22	22	448	480	478	510	0.12	0.13
Odisha	53,799	55,010	39,053	38,781	6,351	7,186	99,204	100,976	25.47	25.20
Sikkim	0	0	58	58	43	43	101	101	0.03	0.03
Telangana	11,257	11,257	8,497	8,627	3,452	3,405	23,206	23,289	5.96	5.81
Uttar Pradesh	884	884	178	178	0	0	1,062	1,062	0.27	0.26
West Bengal	18,752	18,752	11,433	11,425	3,773	4,209	33,958	34,386	8.72	8.58
All India Total	212,207	220,412	148,716	149,048	28,498	31,255	389,421	400,715	100	100
Distribution (%)	54.49	55.00	38.19	37.20	7.32	7.80	100	100		

Total may not tally due to rounding off

Source: Ministry of Coal

Table 1.2: Statewise Estimated Reserves of Lignite(As on 1st April 2025)

(in Million Tonnes)

States/ UTs	Proved		Indicated		Inferred		Total		Distribution (%)	
	2024	2025	2024	2025	2024	2025	2024	2025	2024	2025
Gujarat	1,279	1,279	284	284	1,160	1,160	2,722	2,722	5.76	5.75
Jammu & Kashmir	0	0	20	20	7	7	28	28	0.06	0.06
Kerala	0	0	0	0	10	10	10	10	0.02	0.02
Odisha	6	6	0	0	0	0	6	6	0.01	0.01
Puducherry	0	0	406	406	11	11	417	417	0.88	0.88
Rajasthan	1,204	1,204	3,109	3,183	2,274	2,274	6,586	6,661	13.93	14.06
Tamil Nadu	5,476	5,476	21,412	21,412	10,635	10,635	37,524	37,524	79.34	79.21
West Bengal	0	0	1	1	3	3	4	4	0.01	0.01
All India	7,964	7,964	25,231	25,306	14,100	14,100	47,296	47,371	100	100
Distribution (%)	16.84	16.81	53.35	53.42	29.81	29.76	100	100		

Total may not tally due to rounding off

Source: Ministry of Coal

Chapter 1: Energy Reserves and Potential

Table 1.3: Statewise Estimated Reserves of Crude Oil and Natural Gas (As on 1st April 2025)

States/ UTs/ Region	Crude Oil (Million Tonnes)				Natural Gas (Billion Cubic Metres)			
	2024		2025		2024		2025	
	Estimated Reserves	Distribution (%)	Estimated Reserves	Distribution (%)	Estimated Reserves	Distribution (%)	Estimated Reserves	Distribution (%)
Andhra Pradesh	7.69	1.15	8.41	1.25	59.27	5.42	60.60	5.65
Arunachal Pradesh	2.88	0.43	2.97	0.44	7.33	0.67	7.10	0.66
Assam	145.41	21.66	144.73	21.53	164.51	15.03	163.33	15.22
Gujarat	118.86	17.70	121.20	18.03	55.46	5.07	54.74	5.10
Nagaland	2.38	0.35	2.38	0.35	0.09	0.01	0.09	0.01
Rajasthan	131.50	19.59	136.46	20.30	63.55	5.81	27.12	2.53
Tamil Nadu	8.54	1.27	9.11	1.36	36.57	3.34	38.39	3.58
Tripura	0.07	0.01	0.07	0.01	28.18	2.58	27.73	2.58
West Bengal	0.15	0.02	0.20	0.03	0.00	0.00	0.79	0.07
West Bengal (CBM)	0.00	0.00	0.00	0.00	44.37	4.06	44.93	4.19
Jharkhand (CBM)	0.00	0.00	0.00	0.00	4.86	0.44	5.67	0.53
Madhya Pradesh (CBM)	0.00	0.00	0.00	0.00	24.38	2.23	23.15	2.16
Eastern Offshore	40.67	6.06	37.25	5.54	263.34	24.07	270.67	25.23
Western Offshore	213.26	31.76	209.30	31.14	342.29	31.28	348.70	32.50
Total	671.40	100.00	672.07	100.00	1094.19	100.00	1073.01	100.00
CBM : Cold Bed Methane (Jharkhand, West Bengal and M.P.)								
1. Western offshore includes Gujarat offshore								
2. Total may not tally due to rounding off								
Source: M/o Petroleum & Natural Gas								

Chapter 1: Energy Reserves and Potential

Table 1.4: Sourcewise and Statewise Estimated Potential of Renewable Power in India (as on 31.03.2025)

(in MW)

Sl. No.	States/ UTs	Wind Power @ 150m	Small Hydro Power*	Biomass Power	Cogeneration-bagasse	Solar Energy	Large Hydro#	Total	Distribution (%)
1	Andhra Pradesh	123336	409	1999	280	299312	2596	427933	9.10
2	Arunachal Pradesh	246	2065	18	0	468	50394	53191	1.13
3	Assam	459	202	322	0	19173	643	20799	0.44
4	Bihar	4023	527	964	347	32991	130	38983	0.83
5	Chhattisgarh	2749	1098	354	0	126484	1311	131996	2.81
6	Goa	14	5	33	0	6752	0	6804	0.14
7	Gujarat	180790	202	2638	555	243220	550	427954	9.10
8	Haryana	593	107	1353	362	6468	0	8884	0.19
9	Himachal Pradesh	239	3460	70	0	21502	18305	43576	0.93
10	J& K UT (including Ladakh)	1	1707	83	0	17146	12972	31908	0.68
11	Jharkhand	16	228	146	0	51831	300	52522	1.12
12	Karnataka	169251	3726	1794	1762	223279	4414	404227	8.59
13	Kerala	2621	647	778	0	12405	2473	18924	0.40
14	Madhya Pradesh	55423	820	2516	0	318972	2819	380551	8.09
15	Maharashtra	173868	786	2630	3917	486679	3144	671024	14.26
16	Manipur	0	100	62	0	2294	615	3071	0.07
17	Meghalaya	55	230	69	0	14674	2026	17054	0.36
18	Mizoram	0	169	3	0	612	1927	2711	0.06
19	Nagaland	0	182	54	0	191	325	752	0.02
20	Odisha	12129	286	299	0	139474	2825	155013	3.30
21	Punjab	428	578	3022	414	9210	1301	14954	0.32
22	Rajasthan	284250	52	1300	0	828781	411	1114794	23.70
23	Sikkim	0	267	5	0	254	6051	6577	0.14
24	Tamil Nadu	95107	604	1560	639	204765	1785	304461	6.47
25	Telangana	54717	102	1678	117	140451	1302	198368	4.22
26	Tripura	0	47	34	0	9106	0	9187	0.20
27	Uttar Pradesh	510	461	2800	4926	97843	502	107041	2.28
28	Uttarakhand	49	1664	93	215	4436	13481	19939	0.42
29	West Bengal	1281	392	1742	0	22742	809	26966	0.57
30	Andaman & Nicobar	1245	7	18	0	594	0	1865	0.04
31	Chandigarh	0	0	0	0	22	0	23	0.00
32	Dadar & Nagar Haveli, Daman & Diu	17	0	2	0	498	0	517	0.01
33	Delhi	0	0	0	0	550	0	550	0.01
34	Lakshadweep	31	0	1	0	0	0	32	0.00
35	Puducherry	408	0	5	0	196	0	609	0.01
36	Others\$	0	0	0	284	0	0	284	0.01
All India Total		1,163,856	21,134	28,447	13,818	3,343,378	133,410	4,704,043	100
Distribution (%)		24.74	0.45	0.60	0.29	71.07	2.84	100.00	

\$ Others includes installations through NGOs/IREDA in different states

*capacity upto 25 MW, # capacity > 25 MW

Source: Ministry of New and Renewable Energy